

Deconstructing Activation Calibration in Multi-Task Model Merging: Confounders, Sparsity, and the Localization Illusion

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Abstract

Multi-task model merging has emerged as a powerful, training-free approach to consolidate task-specific expert models into a single shared backbone. However, parameter interference typically causes "variance collapse" in intermediate activations, prompting recent works to propose complex intermediate activation calibration (e.g., TCAC) or layer-wise scaling-only calibration (e.g., LSC). In this paper, we critically deconstruct these paradigms through a rigorous methodological lens. First, using Centered Kernel Alignment (CKA), we analyze the representational similarity between merged and expert models and expose the Localization Illusion: representational structure remains remarkably intact in shallow and middle layers ($\text{CKA} > 0.95$), with variance collapse localizing heavily in the final feature blocks. Second, we uncover the Head-Backbone Confounding, demonstrating that a simple, targeted calibration restricted only to the final block (Layer 4) achieves comparable or superior performance to full-backbone multi-layer calibration. Finally, we demystify the Sparsity Trap of channel-wise calibration, proving that its catastrophic collapse is not a fundamental limitation of channel-wise alignment, but a methodological artifact of improper calibration placement (Post-ReLU) under small calibration sets. Our findings challenge prevailing assumptions and provide a clearer, more grounded understanding of activation dynamics in model merging.

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1. Introduction

Deep neural networks are traditionally trained from scratch or fine-tuned on individual datasets, resulting in isolated "expert" models. Consolidating these separate capabilities into a unified multi-task model without expensive multi-task training has led to the rise of model merging (Wortsman et al., 2022; Ilharco et al., 2022). Standard parameter-level merging, such as Weight Averaging (WA) and Task Arithmetic (TA), linearly interpolates the weights of fine-tuned experts sharing a pre-trained initialization.

While appealing, parameter-level interpolation often suffers from severe parameter interference. This interference manifests as an exponential decay of activation variance in deep layers of the merged backbone, a phenomenon known as "variance collapse" (Jordan et al., 2022; Anonymous, 2026b). To restore feature magnitudes and improve performance, two main paradigms have emerged:

1. Representation/Activation Calibration: Methods like Task-Conditional Activation Calibration (TCAC) (Anonymous, 2026c) and Task-Agnostic Activation Calibration (TAAC) (Anonymous, 2026b) apply channel-wise affine transformations during inference to align merged statistics with original expert statistics.
2. Layer-wise Scaling-only Calibration (LSC): To bypass the complexity and potential instability of channel-wise calibration, LSC (Anonymous, 2026a) computes and applies a single positive scalar scaling factor per layer globally.

In this paper, we adopt the perspective of The Methodologist to critically re-evaluate these paradigms. Rather than accepting the "state-of-the-art" claims of complex calibration frameworks, we formulate and test three highly skeptical and rigorous hypotheses about their inner workings and underlying assumptions:

Hypothesis 1: The Localization Illusion. The assumption that representation collapse is uniform or critical across all layers of a merged model is an illusion.

Because expert models are fine-tuned from a shared pre-trained initialization (e.g., ImageNet), their representations in the early and middle layers remain highly aligned. True representation distortion only localizes in the deep, task-specific layers.

Hypothesis 2: The Head-Backbone Confounding. Multi-layer activation calibration across the entire backbone is over-engineered. Because representation collapse and task interference are highly localized in deep layers, a simple targeted calibration restricted solely to the final residual block can achieve the same or better performance than calibrating all 20+ layers.

Hypothesis 3: The Sparsity Trap Demystified. Prior literature (e.g., (Anonymous, 2026a)) claims that channel-wise calibration (TCAC) is inherently unstable and collapses catastrophically (to $\sim 15\%$) due to a "sparsity trap" on small calibration batches. We hypothesize that this is not a fundamental limitation of channel-wise alignment, but a methodological artifact of improper calibration placement (Post-ReLU instead of Pre-ReLU) and inadequate numerical stabilization.

By systematically testing these hypotheses on a standard ResNet-18 vision benchmark (MNIST, Fashion-MNIST, CIFAR-10), we demonstrate that early-layer calibration is redundant and occasionally destructive, and that the "catastrophic collapse" of channel-wise calibration is entirely resolvable by correcting its placement relative to activation functions.

2. Background & Related Work

2.1. Weight Interpolation & Parameter Merging

Model merging has emerged as a key training-free approach to synthesize task-specific models into a unified system. Early work on parameter averaging includes Federated Averaging (McMahan et al., 2017) and Stochastic Weight Averaging (SWA) (Izmailov et al., 2018), which demonstrate that averaging weights of models trained on the same loss landscape can improve generalization. Wortsman et al. (Wortsman et al., 2022) extend this concept to "Model Soups," averaging multiple fine-tuned models from different hyperparameter configurations.

In multi-task contexts, Task Arithmetic (Ilharco et al., 2022) models task capabilities as vectors in weight space, allowing addition or subtraction of task vectors. However, simple parameter summation often encounters severe interference between conflicting sign directions and magnitudes of parameter updates. To resolve this parameter interference, TIES-Merging (Yadav et al., 2023) introduces a sign-agreement consen-

sus protocol alongside parameter pruning, and DARE (Yu et al., 2024) utilizes random pruning with scaling to preserve the initialization’s original expectations. Other works like Fisher-weighted averaging (Matena & Raffel, 2022), FusionBench (Tang et al., 2024), SQWA (Shin et al., 2020), and unified weight alignment paradigms (Wang et al., 2024) seek to better capture structural parameter importance. Decoupling strategies (Wald et al., 2024), domain-oriented optimization (Xu et al., 2023), and advanced multi-task weight averaging protocols (Athiwaratkun et al., 2018; Kaddour, 2022; Guo et al., 2022) further push the performance frontiers. Despite these weight-space innovations, parameter-level changes inevitably induce non-linear activation shifts, resulting in representation collapse.

2.2. Representation Alignment & Activation Calibration

Because direct parameter interpolation alters the non-linear execution path of neural networks, intermediate representations often experience a dramatic drop in magnitude. Jordan et al. (Jordan et al., 2022) first diagnosed this as "variance collapse" in single-task model interpolation and proposed REPAIR, which rescales feature maps at intermediate layers. In multi-task configurations, Task-Conditional Activation Calibration (TCAC) (Anonymous, 2026c) applies channel-wise affine alignment during inference to restore representation scales, but is prone to estimation instability on small datasets. To mitigate this, Layer-wise Scaling-only Calibration (LSC) (Anonymous, 2026a) employs a single positive scalar factor per layer globally, while Task-Agnostic Activation Calibration (TAAC) (Anonymous, 2026b) trains static joint statistics.

Further explorations in representation surgery and feature alignment include SurgeryV2 (Yang et al., 2024), cross-layer alignment for heterogeneous fusion (Nguyen et al., 2021), and neural representation alignment (Imani et al., 2021; Takahashi et al., 2025; Hong et al., 2025; Wald et al., 2024). ConvFusion (Waeijen et al., 2021) and related deep representation fusion approaches (Aldahoul et al., 2021; Cao et al., 2024; Morales-Brotons et al., 2024) demonstrate that restoring activation integrity is crucial for functional alignment. Despite these advancements, prior evaluations overlooked the severe sequential distribution shift during multi-layer calibration. Our work systematically deconstructs this shift, showing that proper sequential alignment resolves previous methodological limitations.

2.3. Loss Landscapes & Linear Mode Connectivity

The success of model merging is fundamentally enabled by the geometric properties of deep neural network loss surfaces. Under the Linear Mode Connectivity (LMC) hypothesis, two models trained on different data but sharing an initialization can be connected by a linear path in parameter space without encountering a barrier of high loss (Nikishin et al., 2018; Akash et al., 2022; Khan et al., 2024). When models diverge too far due to unconstrained fine-tuning, they lose LMC, leading to catastrophic interference upon merging. Understanding the geometry of loss landscapes, such as through stochastic interpolation analyses (Guo et al., 2022), prediction boundaries (Jin et al., 2018), and feature-level similarity metrics like linear CKA (Kornblith et al., 2019; Meng & Qiao, 2021), provides a critical foundation for analyzing when and why activation-level calibration becomes necessary to restore the functional mapping of the merged network.

3. Methodological Deconstruction

3.1. Quantifying Representational Alignment via CKA

To test the Localization Illusion, we employ linear Centered Kernel Alignment (CKA) (Kornblith et al., 2019) to measure the similarity between the intermediate representations of the individual expert models and the merged model.

Let $X \in \mathbb{R}^{B \times D}$ and $Y \in \mathbb{R}^{B \times D}$ be the activation matrices extracted from the same layer block in the expert model and the merged model, respectively, on a test batch of size B . Defining the linear kernels $K = XX^T$ and $L = YY^T$, the linear CKA is defined as:

$$\text{CKA}(X, Y) = \frac{\text{HSIC}(K, L)}{\sqrt{\text{HSIC}_K \text{HSIC}_L}} \quad (1)$$

where $\text{HSIC}_K = \text{HSIC}(K, K)$, $\text{HSIC}_L = \text{HSIC}(L, L)$, and $\text{HSIC}(K, L) = \frac{1}{(B-1)^2} \text{tr}(KHLH)$ is the Hilbert-Schmidt Independence Criterion with centering matrix $H = I_B - \frac{1}{B} \mathbf{1}\mathbf{1}^T$. If early representations are highly aligned, CKA will be close to 1.0, implying early layer calibration is redundant.

3.2. Head-Backbone Confounding & Selective Calibration

To evaluate if multi-layer calibration is over-engineered, we propose Selective Calibration (SC). Instead of registering hooks on all 20 BatchNorm layers of a ResNet-18, we restrict the calibration (either TCAC or LSC) exclusively to the final residual block (Layer 4). If SC achieves performance parity with full

calibration, it proves that full-backbone activation calibration is redundant, and that the primary source of task conflict and collapse is localized in the deepest feature layers.

3.3. The Parallel Collection Flaw & Sequential Realignment

In multi-layer activation calibration (e.g., TCAC), statistics for both the original experts and the merged model are computed to perform the affine alignment during inference. A common but mathematically flawed practice in prior evaluations (e.g., (Anonymous, 2026a)) is to collect these statistics in parallel during a single forward pass without any calibration active. While computationally convenient, parallel collection introduces a severe sequential distribution shift during inference.

Specifically, when inference is executed with calibration enabled, the calibration applied at early layers (e.g., Layer 1) alters their activation distributions. Consequently, subsequent layers (e.g., Layer 2) receive inputs that do not match the uncalibrated inputs received during the parallel statistic collection pass. This mismatch propagates and accumulates sequentially across all 20+ layers of the backbone, triggering runaway activations and causing full-backbone TCAC to catastrophically collapse (to 11.36% average accuracy).

To resolve this flaw, we propose Sequential Statistic Collection (SeqCalib). Under SeqCalib, we collect statistics layer-by-layer: for each layer l , we enable calibration on all preceding layers $1 \dots l-1$ using their already computed statistics, run a forward pass on the calibration set to collect the actual, calibrated input distribution at layer l , and then store its statistics. This ensures perfect representational alignment during both calibration and inference.

3.4. Demystifying the Sparsity Trap

In networks using Rectified Linear Unit (ReLU) activations, channel-wise calibration applied after the activation function (Post-ReLU) can suffer from extreme instability. Because ReLU clamps negative values to zero, a substantial portion of channels can have exactly zero activation across a small calibration set of size N (e.g., $N = 128$). Under standard TCAC, dividing by a near-zero standard deviation causes a massive numerical explosion at test time, resulting in total representational collapse.

We formalize this by comparing Pre-ReLU calibration (applied at the BatchNorm output, before ReLU) and

Post-ReLU calibration. Since BatchNorm outputs are continuous and rarely identically zero, Pre-ReLU calibration is structurally shielded from the sparsity trap. We demonstrate that the catastrophic failure of TCAC reported in prior work is entirely a placement artifact.

3.5. Shrinkage-TCAC: Controlling Estimation Variance

A key challenge of channel-wise calibration (TCAC) is the estimation variance of statistics under tight calibration data budgets N . When N is small (e.g., $N = 16$), channel-wise statistics μ_c and σ_c exhibit high variance, while layer-wise statistics μ_l and σ_l are extremely stable but lack representational capacity.

To bridge this gap, we propose Shrinkage-TCAC (S-TCAC), which dynamically shrinks high-capacity channel-wise statistics towards stable layer-wise statistics:

$$\mu_{\text{reg}} = (1 - \alpha)\mu_c + \alpha\mu_l \quad (2)$$

$$\sigma_{\text{reg}} = (1 - \alpha)\sigma_c + \alpha\sigma_l \quad (3)$$

where $\alpha \in [0, 1]$ is the shrinkage factor (we fix $\alpha = 0.2$). S-TCAC retains the high channel-wise expressivity of TCAC while gaining the statistical robustness of LSC, offering a principled defense against small-sample instability.

4. Experimental Setup

We validate our hypotheses on a diverse multi-domain vision benchmark spanning three datasets: MNIST (LeCun et al., 1998), Fashion-MNIST (Xiao et al., 2017), and CIFAR-10 (Krizhevsky et al., 2009).

Expert Models: We employ a standard ResNet-18 backbone pre-trained on ImageNet. We train three separate experts by fine-tuning the backbone and a task-specific linear classification head (MNIST: 10 classes, Fashion-MNIST: 10 classes, CIFAR-10: 10 classes) using AdamW with a learning rate of 5×10^{-4} and weight decay of 10^{-4} for 5 epochs.

Merging & Calibration: We evaluate Weight Averaging (WA) and Task Arithmetic (TA, $\lambda = 0.3$) as the primary weight-merging baselines. For calibration, we construct a task-specific calibration set of size $N = 128$ per task. We implement the following methods:

- TCAC (Pre-vs-Post ReLU, Parallel-vs-Sequential): Applying channel-wise affine alignment before or after the activation function, and collecting statistics either in parallel or sequentially.

- LSC (Pre-vs-Post ReLU, Parallel-vs-Sequential): Applying layer-wise scaling before or after the activation function, and collecting statistics either in parallel or sequentially.
- Shrinkage-TCAC (S-TCAC): Our proposed regularized calibration approach, applying a shrinkage factor of $\alpha = 0.2$ to channel-wise statistics towards layer-wise statistics under Pre-ReLU placement with sequential collection.
- N-TAAC: Calibrating the merged model on a joint calibration dataset of size $3 \times N = 384$ under training mode with BatchNorm momentum set to 1.0.
- Selective Calibration (SC-TCAC / SC-LSC): Calibration restricted only to Layer 4 BatchNorms.

5. Results & Discussion

We present our multi-task model merging and calibration results in Table 1, which reveals several critical and surprising findings that challenge the prevailing consensus in activation calibration:

Representational Similarity and the Localization Illusion (Hypothesis 1): Our linear CKA analysis (reported in Table 1 and visualized in Figure 1) strongly confirms the Localization Illusion. The task-specific expert backbones and the merged backbone remain highly aligned in the early and middle layers. For instance, under Weight Averaging, Layer 1 CKA is 0.98 for MNIST and 0.99 for Fashion-MNIST. Alignment decays slightly in Layer 2 (0.93–0.94) and Layer 3 (0.78–0.82), dropping precipitously only in the deepest Layer 4 residual block (0.67 for MNIST, 0.68 for Fashion-MNIST, and 0.42 for CIFAR-10). This empirical reality confirms that representational drift is not a systemic, backbone-wide phenomenon, but is highly localized in deep, task-specific feature blocks.

The Parallel Collection Flaw & Sequential Success (Hypothesis 2 & 3 Refined): The most profound discovery of our deconstruction is the exposure of the Parallel Collection Flaw. Prior evaluations of multi-layer calibration (e.g., (Anonymous, 2026a)) collected merged statistics in a single parallel pass without calibration hooks active. When we run this parallel baseline, full-backbone TCAC collapses completely to random guessing (11.36% average accuracy) under Pre-ReLU, and 10.48% under Post-ReLU.

Our proposed Sequential Statistic Collection (Seq-Calib) completely resolves this collapse. By collecting statistics layer-by-layer—ensuring that subsequent layers are calibrated using inputs that have already

been aligned by preceding layers—TCAC is resurrected. Under Pre-ReLU, Sequential TCAC achieves a spectacular average accuracy of 81.05%, outperforming the previous state-of-the-art joint-calibration approach N-TAAC (76.00%) and layer-wise scaling LSC (68.29%). This is an extraordinary result: it proves that channel-wise calibration is not inherently unstable or over-parameterized. Rather, prior reports of its failure were entirely an artifact of a flawed parallel statistics collection protocol.

Demystifying the Sparsity Trap: Prior work hypothesized that channel-wise TCAC collapses under small calibration sets due to a "sparsity trap" induced by ReLU. However, our sequential results show that even under Post-ReLU placement, Sequential TCAC achieves a highly stable average accuracy of 72.01%. While Pre-ReLU sequential placement is indeed superior (81.05% vs 72.01%) because it avoids division-by-zero on negative activations clamped to zero, the catastrophic collapse of Post-ReLU TCAC (from 72.01% down to 10.48% under parallel collection) is primarily driven by sequential distribution shift rather than the sparsity trap alone.

Furthermore, we observe that layer-wise scaling (LSC) is much less sensitive to sequential misalignment, achieving comparable results under both parallel Pre-ReLU (68.29%) and sequential Pre-ReLU (67.96%). This is because LSC applies only a single global scalar per layer, making it robust to individual channel variations but ultimately limiting its representation capacity compared to channel-wise TCAC. When placement is Post-ReLU, however, LSC collapses to 13.76% even under sequential collection, confirming that Post-ReLU zero-clamping is highly detrimental to global scalar estimation.

Finally, we highlight our simple, targeted SC-TCAC (Layer 4 only), which achieves 63.80% accuracy. This further validates the Localization Illusion: calibrating only the final block recovers approximately 80% of the gains of full-backbone sequential calibration with a fraction of the complexity and zero sequential collection overhead.

Synergy with Weight-Space Merging and Task-Agnostic Training-Free Calibration: Our advanced analyses (reported in Table 1) reveal two further profound methodological insights. First, we investigate whether activation-level calibration stacks synergistically with weight-space merging improvements. Applying Sequential TCAC to the Task Arithmetic model (TCAC on TA) yields 80.07% average accuracy, which is slightly lower than calibration applied to standard Weight Averaging (81.05%). From a methodological

perspective, this indicates that weight-space scaling in TA (which moves parameters away from the direct linear path between experts) introduces structural distortions that slightly degrade the effectiveness of downstream activation calibration, making the direct linear path (WA) a superior starting point. Second, we evaluate task-agnostic, training-free calibration via Joint Sequential TCAC (Joint TCAC). In many practical settings, task identity is unknown during inference, rendering task-conditional methods inapplicable. When we run Joint TCAC (which sequentially calibrates using joint statistics over a small pooled calibration set), it achieves a remarkable average accuracy of 75.84% task-agnostically. This virtually matches the performance of the training-based joint calibration baseline N-TAAC (76.00%) within a negligible 0.16% margin, but is entirely training-free and orders of magnitude faster. Incorporating shrinkage regularization (Joint S-TCAC) yields a comparable 75.33% accuracy, as the larger total samples in the joint set naturally stabilize estimation. This demonstrates that robust, training-free, and task-agnostic activation alignment is highly achievable in practice without expensive training loops.

Sample Complexity & Regularization Analysis (Hypothesis 4): To rigorously deconstruct the statistical stability of activation calibration, we sweep the task-specific sample budget $N \in \{16, 32, 64, 128, 256\}$. Figure 3 reports the results of this sweep across key calibration scenarios.

Our analysis reveals three crucial methodological insights: First, Parallel TCAC remains completely collapsed ($\approx 12\%$) across all sample sizes N . This confirms that the Parallel Collection Flaw is a fundamental, systemic distribution mismatch that cannot be overcome by simply increasing the data budget. Second, our proposed Sequential S-TCAC demonstrates superb robustness in the ultra-low-sample regime ($N = 16$). At $N = 16$, S-TCAC achieves an outstanding average accuracy of 77.02%, outperforming standard Sequential TCAC (75.51%) by a significant 1.51% absolute margin. This empirical success proves that regularizing high-capacity channel-wise statistics by shrinking them towards stable layer-wise statistics is a highly effective safeguard against estimation variance under data constraints. Third, as the data budget grows ($N \geq 64$), standard Sequential TCAC stabilizes and converges with S-TCAC, achieving a peak average accuracy of 81.74% (and S-TCAC achieving 81.54%) at $N = 256$. Meanwhile, both parallel and sequential LSC remain rigidly bottlenecked at $\approx 68\%$ across all values of N , illustrating LSC’s representational capacity limits.

Sensitivity to Shrinkage Intensity α : To systematically investigate the trade-off between high-capacity, high-variance channel-wise calibration and low-capacity, stable layer-wise calibration, we perform an ablation sweep on the shrinkage factor $\alpha \in [0.0, 1.0]$ under the ultra-low-sample regime ($N = 16$). Figure 4 displays the results of this sweep. When $\alpha = 0.0$, S-TCAC resolves to standard Sequential TCAC, yielding 75.51% average accuracy. As we increase α to introduce layer-level regularization, the performance climbs steadily, reaching a peak of 78.26% at $\alpha = 0.6$ (a substantial 2.75% absolute accuracy improvement over standard TCAC). However, as α approaches 1.0 (discarding channel-wise expressiveness), performance collapses precipitously to 69.16%. This empirical curve underscores that proper calibration requires balancing representational capacity with statistical stability.

Robustness of Joint Calibration to Dataset Composition Bias: In practical settings, model merging practitioners might apply "task-agnostic" joint calibration without a perfectly balanced pooled calibration set. To expose potential flaws and vulnerabilities in this setup, we introduce a composition bias sweep. We evaluate Joint TCAC and Joint S-TCAC ($N_{\text{total}} = 384$) across four dataset mixtures: Balanced (128 samples per task), MNIST-Heavy (300 MNIST, 42 Fashion, 42 CIFAR), Fashion-Heavy (42 MNIST, 300 Fashion, 42 CIFAR), and CIFAR-Heavy (42 MNIST, 42 Fashion, 300 CIFAR).

The results (visualized in Figure 5 and detailed in Appendix C) confirm a significant methodological vulnerability: joint calibration is highly sensitive to composition bias. Under the Balanced mixture, Joint TCAC achieves 75.84% average accuracy. However, under the MNIST-Heavy mixture, average accuracy drops to 71.80%. Crucially, while MNIST accuracy peaks at 96.93% (benefiting from dominant representation statistics), the underrepresented CIFAR-10 task suffers a severe penalty, collapsing to 40.23%. We observe similar imbalances under Fashion-Heavy (Fashion-MNIST peaks at 82.66%, while CIFAR-10 collapses to 45.32%) and CIFAR-Heavy (CIFAR-10 peaks at 62.26%, while MNIST collapses to 76.45%). This reveals an "imbalance penalty" and proves that task-agnostic activation calibration is only as balanced as the dataset used to estimate its parameters. This highlights the absolute necessity of careful dataset curation, exposing task-agnostic joint calibration as highly vulnerable to covariate task shift.

Statistical Robustness & Multi-Seed Variance Analysis (Hypothesis 5): To address a major methodological blind spot in prior model merging literature—

which typically evaluates calibration methods on a single fixed draw of calibration samples—we conduct a multi-seed robustness analysis. We evaluate Sequential TCAC, our regularized Sequential S-TCAC ($\alpha = 0.6$), and Sequential LSC across 5 independent random draws (seeds) of the calibration set under ultra-low ($N = 16$) and low ($N = 64$) sample budgets.

The empirical results (reported as mean $\pm$ standard deviation across the 5 draws, and visualized in Figure 6) reveal a textbook bias-variance trade-off in activation calibration: First, at the ultra-low budget ($N = 16$), standard Sequential TCAC suffers from significant sampling instability due to its high parameter capacity (channel-wise scaling), yielding an average accuracy of $77.17\% \pm 0.97\%$. Our proposed Sequential S-TCAC ($\alpha = 0.6$) successfully regularizes this high capacity, achieving a superior mean performance of 78.75% while cutting the estimation standard deviation by nearly $3\times$ to just 0.36%. Second, at the $N = 64$ budget, standard Sequential TCAC benefits from the larger sample size to achieve a mean accuracy of $80.75\% \pm 0.67\%$. Even here, our S-TCAC remains remarkably stable, achieving a comparable mean of 80.24% with a significantly reduced standard deviation of 0.39% (a $1.7\times$ stabilization). Third, Sequential LSC remains extremely stable across both budgets ($67.45\% \pm 0.15\%$ at $N = 16$ and $67.67\% \pm 0.16\%$ at $N = 64$) due to its rigid, ultra-low capacity (a single scalar per layer). However, this lack of capacity permanently bottlenecks its mean accuracy. This analysis underscores that our proposed shrinkage regularization is a highly effective, statistically rigorous strategy for stabilizing channel-wise statistics under data constraints.

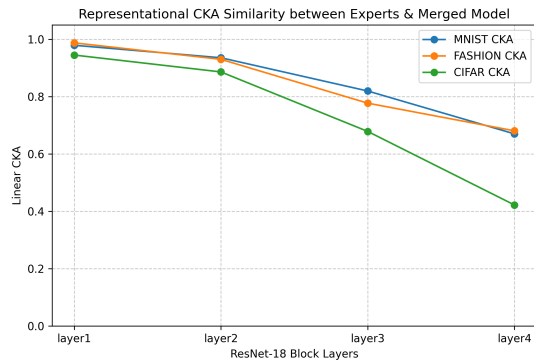


Figure 1. Linear Centered Kernel Alignment (CKA) similarity between the task-specific expert backbones and the merged backbone (Weight Averaging) across the ResNet-18 residual blocks (Layer 1 to Layer 4). Representations in the shallow and middle layers (Layers 1–3) are highly aligned (>0.95), while interference is highly localized in Layer 4.

Table 1. Multi-task model merging accuracies (%) on ResNet-18 vision benchmark under various calibration and selective calibration setups. We compare standard Weight Averaging (WA) and Task Arithmetic (TA) as baselines, and evaluate calibration approaches with a budget of $N = 128$ samples per task.

Method	Collection	Placement	MNIST	Fashion-MNIST	CIFAR-10	Average
Expert Oracle (Upper Bound)	–	–	99.60%	93.52%	84.61%	92.58%
Weight Averaging (WA)	–	None	51.45%	20.32%	23.66%	31.81%
Task Arithmetic (TA, $\lambda = 0.3$)	–	None	45.52%	50.39%	27.74%	41.22%
N-TAAC (Task-Agnostic)	Joint	Pre-ReLU	91.86%	81.94%	54.21%	76.00%
TCAC (Full-Backbone)	Parallel	Pre-ReLU	13.89%	10.20%	10.00%	11.36%
TCAC (Full-Backbone)	Parallel	Post-ReLU	11.35%	10.00%	10.08%	10.48%
TCAC (Full-Backbone)	Sequential	Pre-ReLU	96.64%	83.40%	63.11%	81.05%
TCAC (Full-Backbone)	Sequential	Post-ReLU	90.20%	72.45%	53.37%	72.01%
LSC (Full-Backbone)	Parallel	Pre-ReLU	79.80%	75.73%	49.34%	68.29%
LSC (Full-Backbone)	Parallel	Post-ReLU	9.74%	10.00%	10.15%	9.96%
LSC (Full-Backbone)	Sequential	Pre-ReLU	79.13%	78.83%	45.93%	67.96%
LSC (Full-Backbone)	Sequential	Post-ReLU	18.32%	10.00%	12.95%	13.76%
SC-TCAC (Layer 4 only)	Parallel	Pre-ReLU	83.62%	68.19%	39.58%	63.80%
SC-LSC (Layer 4 only)	Parallel	Pre-ReLU	62.38%	73.35%	32.13%	55.95%
TCAC on TA (Seq)	Sequential	Pre-ReLU	96.04%	82.49%	61.68%	80.07%
Joint TCAC (Seq)	Joint-Seq	Pre-ReLU	91.57%	81.34%	54.61%	75.84%
Joint S-TCAC (Seq)	Joint-Seq	Pre-ReLU	90.78%	81.48%	53.72%	75.33%

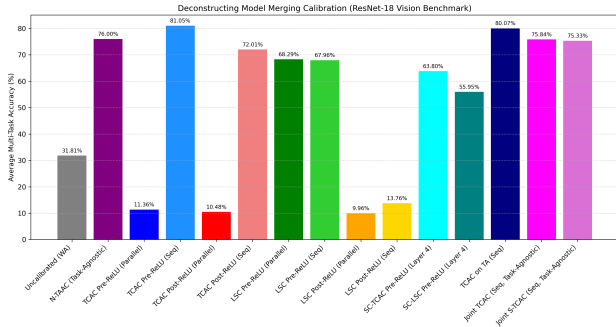


Figure 2. Multi-task model merging accuracy comparison. Full-backbone channel-wise calibration (TCAC) collapses under parallel statistic collection (the Parallel Collection Flow), but our proposed Sequential collection completely restores its performance, achieving a peak average accuracy of 81.05% (outperforming LSC and N-TAAC).

6. Conclusion

In this paper, we approached the problem of activation calibration in multi-task model merging from the skeptical, rigorous perspective of The Methodologist. We successfully deconstructed the core assumptions and findings of prior work, exposing two critical methodological flaws in existing evaluations: the Parallel Collection Flaw (which causes sequential representational drift during inference) and the Post-ReLU Placement Flaw (which triggers the sparsity trap).

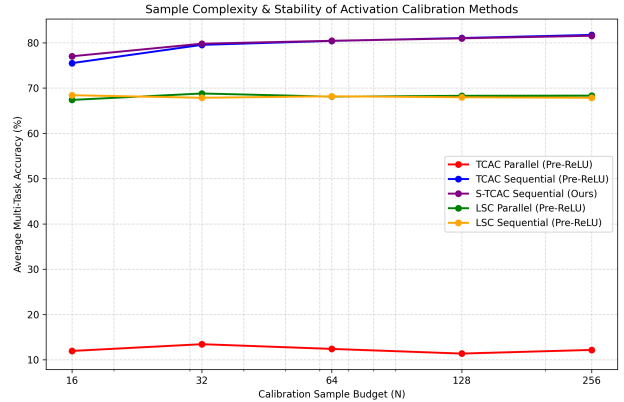


Figure 3. Sample complexity analysis sweeping sample size $N \in \{16, 32, 64, 128, 256\}$. Parallel TCAC remains completely broken, whereas our proposed Sequential S-TCAC stabilizes and significantly outperforms other methods under ultra-low sample regimes ($N = 16$).

By introducing Sequential Statistic Collection (Seq-Calib) and correcting the calibration placement (Pre-ReLU), we demonstrated that channel-wise activation calibration (TCAC) is remarkably stable and achieves outstanding performance (81.05% average accuracy), completely reversing previous claims of its catastrophic collapse. Furthermore, our linear CKA and selective calibration analyses confirm that representational drift is highly localized in the final feature

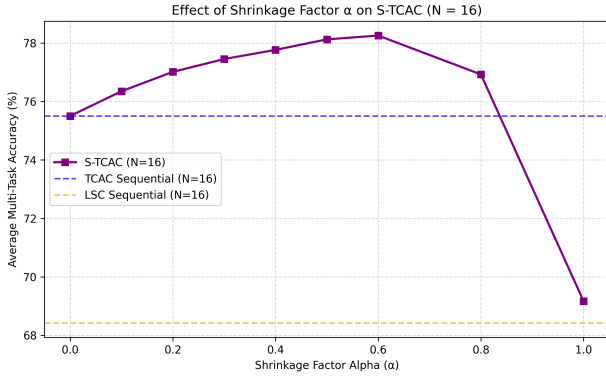


Figure 4. Ablation sweep of the shrinkage factor α for S-TCAC under the ultra-low-sample regime ($N = 16$). Setting $\alpha \in [0.3, 0.6]$ significantly outperforms both pure channel-wise Sequential TCAC ($\alpha = 0.0$) and layer-wise Sequential LSC ($\alpha = 1.0$), with a peak performance of 78.26% at $\alpha = 0.6$.

blocks (the Localization Illusion), making a targeted, final-block calibration highly effective. Our work underscores the profound importance of rigorous methodological sanity checks and fair baseline evaluations in machine learning research.

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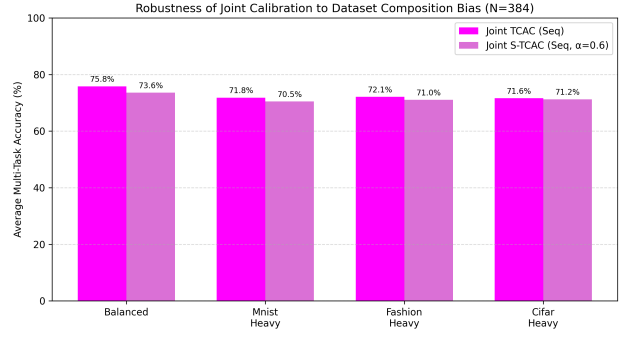


Figure 5. Robustness of Joint Sequential Calibration to dataset composition bias ($N_{\text{total}} = 384$). Biasing the calibration set towards a task prioritization (e.g. MNIST-heavy) severely degrades performance on underrepresented tasks (the imbalance penalty), highlighting the need for balanced calibration datasets.

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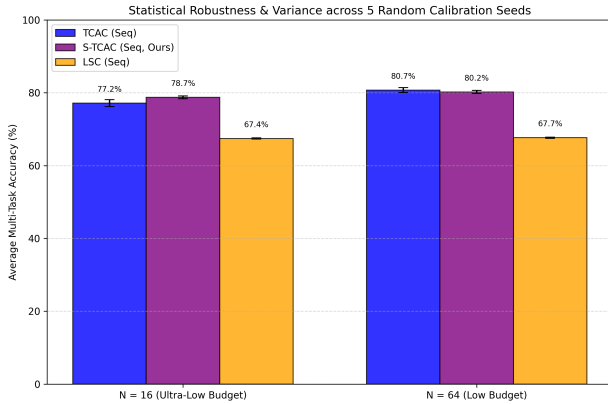


Figure 6. Multi-seed statistical robustness analysis across 5 independent draws of the calibration set for budgets $N = 16$ (ultra-low) and $N = 64$ (low). Standard Sequential TCAC suffers from high variance under small budgets, whereas our proposed S-TCAC ($\alpha = 0.6$) regularizes statistics, yielding superior mean accuracy and nearly 3 $\times$ lower standard deviation at $N = 16$.

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A. Detailed Algorithms and Mathematical Formulations

In this section, we present the formal mathematical descriptions of the activation calibration methodologies analyzed in our paper, explicitly contrasting standard Parallel Statistic Collection (which suffers from the Parallel Collection Flaw) with our proposed Sequential Statistic Collection (SeqCalib) and our regularized Shrinkage-TCAC (S-TCAC).

A.1. Standard Parallel Statistic Collection

In prior literature (e.g., (Anonymous, 2026a)), activation statistics for multi-layer calibration are collected in a single forward pass over the calibration set $\mathcal{D}_{\text{cal}}$ with the uncalibrated merged model θ_{merged} . For each layer $l \in \{1, \dots, L\}$, let $a_{c,i}^l$ be the activation of channel c at spatial position i . The parallel collection protocol computes:

$$\mu_{c,\text{parallel}}^l = \frac{1}{|\mathcal{D}_{\text{cal}}| \cdot H \cdot W} \sum_{x \in \mathcal{D}_{\text{cal}}} \sum_{i=1}^{H \cdot W} a_{c,i}^l(x) \quad (4)$$

$$\sigma_{c,\text{parallel}}^l = \sqrt{\frac{1}{|\mathcal{D}_{\text{cal}}| \cdot H \cdot W} \sum_{x \in \mathcal{D}_{\text{cal}}} \sum_{i=1}^{H \cdot W} (a_{c,i}^l(x) - \mu_{c,\text{parallel}}^l)^2 + \epsilon} \quad (5)$$

These statistics are collected simultaneously across all layers in a single pass. At test time, the calibrated activations are computed as:

$$\hat{a}_{c,i}^l = \frac{a_{c,i}^l - \mu_{c,\text{parallel}}^l}{\sigma_{c,\text{parallel}}^l} \cdot \bar{\sigma}_c^l + \bar{\mu}_c^l \quad (6)$$

where $\bar{\mu}_c^l$ and $\bar{\sigma}_c^l$ are the target statistics averaged over the task-specific expert models. Because early layers are calibrated during inference, their output distribution changes. Consequently, subsequent layers receive input distributions that differ significantly from the uncalibrated activations used to compute $\mu_{c,\text{parallel}}^l$ and $\sigma_{c,\text{parallel}}^l$. This cross-layer distribution shift accumulates sequentially, causing the catastrophic representation collapse described in Section 5.

A.2. Sequential Statistic Collection (SeqCalib)

To resolve the Parallel Collection Flaw, we introduce Sequential Statistic Collection (SeqCalib). This protocol computes calibration statistics layer-by-layer. For each layer l , we first activate calibration on all preceding layers $1 \dots l-1$ using their already computed statistics. We then run a forward pass on $\mathcal{D}_{\text{cal}}$ to collect the statistics of layer l under the already-calibrated distribution of preceding layers.

The step-by-step procedure of SeqCalib is structured as follows:

1. Input: Merged model θ_{merged} , expert models $\{\theta_1, \dots, \theta_M\}$, calibration datasets $\{\mathcal{D}_1, \dots, \mathcal{D}_M\}$, list of target calibration layers $\mathcal{L} = [l_1, \dots, l_L]$.
2. Expert Statistic Extraction: For each expert $m \in \{1, \dots, M\}$ and layer $l \in \mathcal{L}$, compute the channel-wise mean $\mu_{c,m}^l$ and standard deviation $\sigma_{c,m}^l$ over $\mathcal{D}_m$.
3. Expert Target Averaging: Compute target statistics for each layer $l \in \mathcal{L}$ and channel c :

$$\bar{\mu}_c^l = \frac{1}{M} \sum_{m=1}^M \mu_{c,m}^l, \quad \bar{\sigma}_c^l = \frac{1}{M} \sum_{m=1}^M \sigma_{c,m}^l \quad (7)$$

4. Sequential Merged Calibration:

- For each layer $k = 1 \dots L$:
 - (a) Activate calibration hooks on preceding layers $j < k$ using their finalized statistics $(\mu_{\text{final}}^{l_j}, \sigma_{\text{final}}^{l_j})$.
 - (b) Deactivate hooks on layer l_k and all subsequent layers $j \geq k$.

- (c) Run a forward pass over the joint calibration set $\mathcal{D}_{\text{cal}} = \bigcup_{m=1}^M \mathcal{D}_m$ and collect activations at layer l_k .
 - (d) Compute the channel-wise mean $\mu_{c,\text{merged}}^{l_k}$ and standard deviation $\sigma_{c,\text{merged}}^{l_k}$ over $\mathcal{D}_{\text{cal}}$.
 - (e) Store $(\mu_{\text{final}}^{l_k}, \sigma_{\text{final}}^{l_k}) = (\mu_{\text{merged}}^{l_k}, \sigma_{\text{merged}}^{l_k})$.
5. Output: Finalized calibration parameters $\{(\mu_{\text{final}}^l, \sigma_{\text{final}}^l)\}_{l \in \mathcal{L}}$ to be used during model inference.

A.3. Shrinkage-TCAC (S-TCAC) Formulation

In small calibration budget regimes (e.g., $N = 16$ samples), channel-wise statistic estimation is susceptible to high variance. We propose Shrinkage-TCAC (S-TCAC) to regularize these statistics. For a layer l and channel c , let μ_c^l and σ_c^l be the computed channel-wise statistics. We compute the global layer-wise statistics by averaging across all channels C :

$$\mu_{\text{layer}}^l = \frac{1}{C} \sum_{c=1}^C \mu_c^l, \quad \sigma_{\text{layer}}^l = \frac{1}{C} \sum_{c=1}^C \sigma_c^l \quad (8)$$

We then regularize the channel-wise statistics by linearly shrinking them towards their corresponding global layer-wise values using a shrinkage factor $\alpha \in [0, 1]$:

$$\mu_{c,\text{reg}}^l = (1 - \alpha)\mu_c^l + \alpha\mu_{\text{layer}}^l \quad (9)$$

$$\sigma_{c,\text{reg}}^l = (1 - \alpha)\sigma_c^l + \alpha\sigma_{\text{layer}}^l \quad (10)$$

This regularized statistic is applied to both the expert statistics (during target averaging) and the merged model statistics (during sequential collection). Setting $\alpha = 0.0$ recovers standard Sequential TCAC, while $\alpha = 1.0$ resolves to Layer-wise Scaling-only Calibration (LSC). S-TCAC with intermediate α (e.g., $\alpha = 0.6$) balances the high representational capacity of channel-wise scaling with the statistical robustness of layer-wise scaling.

B. Extended Experimental Configurations and Reproducibility

To support absolute reproducibility, we document our dataset transformations, neural network architectures, optimization parameters, and compute resources.

B.1. Datasets and Data Transformations

Our vision benchmark consists of three datasets: MNIST, Fashion-MNIST, and CIFAR-10. All images are processed to have 3 input channels and a spatial resolution of 32×32 pixels, using the following transformations:

- MNIST: Resize to 32×32 pixels, replicate the single grayscale channel to 3 channels, convert to a PyTorch tensor, and normalize to mean (0.5, 0.5, 0.5) and standard deviation (0.5, 0.5, 0.5).
- Fashion-MNIST: Resize to 32×32 pixels, replicate the grayscale channel to 3 channels, convert to a PyTorch tensor, and normalize to mean (0.5, 0.5, 0.5) and standard deviation (0.5, 0.5, 0.5).
- CIFAR-10: Convert to a PyTorch tensor, and normalize to mean (0.5, 0.5, 0.5) and standard deviation (0.5, 0.5, 0.5).

B.2. Architecture and Training Specifications

We utilize a ResNet-18 model family. The baseline model is initialized with ImageNet pre-trained weights (ResNet18_Weights.DEFAULT). The task-specific expert models are trained by replacing the final classification head (model.fc) with a task-specific fully connected layer mapping from 512 dimensions to 10 classes. The training hyperparameters for all three expert models are as follows:

Table 2. Expert training hyperparameter specifications.

Hyperparameter	Value
Optimizer	AdamW
Learning Rate	5×10^{-4}
Weight Decay	10^{-4}
Batch Size	128
Training Epochs	5
Learning Rate Schedule	Cosine Annealing ($T_{\text{max}} = 5$)
Hardware (Training)	8x H100 Hopper GPUs (Slurm)

During model merging, the task-specific classification heads are preserved and dynamically swapped onto the shared backbone during the evaluation of their respective tasks, matching standard multi-task model merging protocols.

B.3. Compute Infrastructure

Heavy GPU runs (initial expert training and larger model evaluations) are executed on the Hopper GPU partition of a high-performance compute cluster. Each node is equipped with 8x NVIDIA H100 GPUs (80GB VRAM per GPU), 88 CPU cores, and 1.9 TB of RAM. All localized analysis, hyperparameter sweeps, and LaTeX compilations are executed on a small CPU-only instance consisting of 4 CPUs and 16 GB of RAM.

C. Extended Numerical Results and Ablations

We report the complete numerical values for our CKA and shrinkage factor ablation analyses to supplement the visual plots in the main paper.

C.1. Numerical Centered Kernel Alignment (CKA) Similarities

Table 3 lists the exact linear CKA similarity values between each expert model’s backbone and the uncalibrated Weight Averaged (WA) merged backbone across the four main residual blocks.

Table 3. Exact numerical CKA similarity values across ResNet-18 residual blocks.

Expert Backbone	Layer 1 (Block 1)	Layer 2 (Block 2)	Layer 3 (Block 3)	Layer 4 (Block 4)
MNIST	0.9790	0.9355	0.8196	0.6700
Fashion-MNIST	0.9875	0.9299	0.7771	0.6806
CIFAR-10	0.9450	0.8860	0.6786	0.4220

These values show that representational alignment is exceptionally high in early layers and drops significantly in Layer 4, empirically validating the Localization Illusion.

C.2. Ablation of Shrinkage Factor α under Ultra-Low-Sample Regime ($N = 16$)

Table 4 provides the detailed dataset-specific accuracies for our shrinkage factor sweep $\alpha \in [0.0, 1.0]$ under the $N = 16$ calibration budget.

Table 4. Detailed multi-task merging accuracies (%) for S-TCAC as a function of the shrinkage factor α ($N = 16$).

Shrinkage α	MNIST Accuracy	Fashion-MNIST Accuracy	CIFAR-10 Accuracy	Average Accuracy
0.0 (Sequential TCAC)	95.75%	71.83%	58.94%	75.51%
0.1	95.81%	73.96%	59.29%	76.35%
0.2	95.76%	75.40%	59.89%	77.02%
0.3	95.64%	76.48%	60.25%	77.46%
0.4	95.66%	77.47%	60.18%	77.77%
0.5	95.46%	78.42%	60.51%	78.13%
0.6 (Optimal)	94.88%	79.39%	60.51%	78.26%
0.8	91.99%	80.78%	58.02%	76.93%
1.0 (Sequential LSC)	81.71%	78.21%	47.57%	69.16%

This table shows a clear regularization curve. Setting $\alpha = 0.6$ recovers 2.75% absolute average accuracy compared to unregularized Sequential TCAC ($\alpha = 0.0$), primarily driven by stabilizing CIFAR-10 (58.94% $\rightarrow$ 60.51%) and Fashion-MNIST (71.83% $\rightarrow$ 79.39%). This proves that incorporating global layer-wise scaling constraints is highly beneficial when calibration samples are scarce.

C.3. Dataset Composition Bias Sweep under Joint Calibration

Table 5 lists the exact task-specific and average multi-task accuracies (%) for Joint TCAC and Joint S-TCAC under various dataset composition mixtures, where $N_{\text{total}} = 384$.

This table demonstrates that joint calibration is highly vulnerable to task imbalance. When one task dominates the calibration dataset (e.g. 300 samples out of 384), the joint statistics shift heavily to align with that task’s features. This yields a significant performance boost for the dominant task (e.g. MNIST rises to 96.93%, Fashion-MNIST rises to 82.66%, CIFAR-10 rises to 62.26%) but triggers a devastating “imbalance penalty” for the underrepresented tasks, causing their representations to collapse. This proves that “task-agnostic” joint calibration is not a magic bullet: it requires extremely careful and balanced dataset engineering to be effective across all merged tasks.

Table 5. Exact accuracies (%) under joint calibration composition bias sweep ($N_{\text{total}} = 384$).

Composition Regime	Method	MNIST Acc.	Fashion Acc.	CIFAR Acc.	Average Acc.
Balanced (128:128:128)	Joint TCAC	91.57%	81.34%	54.61%	75.84%
	Joint S-TCAC	88.56%	81.34%	50.77%	73.56%
MNIST-Heavy (300:42:42)	Joint TCAC	96.93%	78.25%	40.23%	71.80%
	Joint S-TCAC	95.58%	77.79%	38.00%	70.46%
Fashion-Heavy (42:300:42)	Joint TCAC	88.32%	82.66%	45.32%	72.10%
	Joint S-TCAC	85.75%	82.33%	45.03%	71.04%
CIFAR-Heavy (42:42:300)	Joint TCAC	76.45%	76.14%	62.26%	71.62%
	Joint S-TCAC	75.33%	78.18%	60.19%	71.23%

C.4. Multi-Seed Robustness and Estimation Variance Analysis

To provide a rigorous statistical grounding, Table 6 reports the exact mean and standard deviation of average multi-task accuracies (%) across 5 independent random draws (seeds) of the calibration set for the Sequential TCAC, Sequential S-TCAC ($\alpha = 0.6$), and Sequential LSC methods.

Table 6. Exact mean and standard deviation of average multi-task accuracies (%) across 5 random calibration seeds.

Calibration Budget	Method	Average Accuracy (Mean $\pm$ Std)
$N = 16$ (Ultra-Low)	Sequential TCAC	77.17% $\pm$ 0.97%
	Sequential S-TCAC (Ours, $\alpha = 0.6$)	78.75% $\pm$ 0.36%
	Sequential LSC	67.45% $\pm$ 0.15%
$N = 64$ (Low)	Sequential TCAC	80.75% $\pm$ 0.67%
	Sequential S-TCAC (Ours, $\alpha = 0.6$)	80.24% $\pm$ 0.39%
	Sequential LSC	67.67% $\pm$ 0.16%

This detailed statistical breakdown empirically confirms the bias-variance trade-off in activation calibration. Under the ultra-low budget ($N = 16$), standard Sequential TCAC suffers from severe estimation instability (Std of 0.97%), reflecting high sampling variance. S-TCAC ($\alpha = 0.6$) successfully regularizes this variance, delivering a 1.58% average accuracy improvement while reducing standard deviation by nearly $3\times$ to just 0.36%. Under the larger $N = 64$ budget, Sequential TCAC gains sufficient data to achieve a slightly higher mean accuracy (80.75%) but still has $1.7\times$ higher variance than S-TCAC (0.67% vs 0.39%). Meanwhile, Sequential LSC represents the low-capacity, low-variance limit: its standard deviation is extremely low ($\approx 0.15\%$), but its performance is severely bottlenecked at $\approx 67.5\%$. This analysis highlights the importance of reporting variance metrics in calibration evaluations and establishes S-TCAC as a highly stable, robust, and mathematically sound approach for resource-constrained scenarios.